

SIMATS ENGINEERING
ASSESSMENT TOOL 2: LABORATORY PERFORMANCE

COURSE NAME:	Programming in Java	COURSE CODE:	CSA0915- SLOT B
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ASSESSMENT TASKS:

3. Develop an AWT-based Student Registration Form using: Label

TextField

CheckboxGroup

Choice

Button

Frame- Arrange the components using an appropriate layout manager. " Test

Cases:

tc1_input- Student ID=S101, Name=Ravi Kumar, Gender=Male,

Department=Information Technology tc1_output- Student

Registered Successfully

tc2_input- Student ID=S102, Name=Priya, Gender=Female, Department=Computer

Science tc2_output- Registration Successful and student details displayed

tc3_input- Student Name is empty

tc3_output- Student Name is required

CODE:

A. JAVA CODE import

```
java.awt.*;      import
```

```
java.awt.event.*;
```

```
public class StudentRegistration extends Frame implements ActionListener {
```

```

Label idLabel, nameLabel, genderLabel, deptLabel;
TextField idField, nameField;
CheckboxGroup genderGroup;
Checkbox male, female;
Choice department;
Button registerButton, clearButton;
TextArea resultArea;

public StudentRegistration() {

    setTitle("Student Registration Form");
setSize(600, 450);    setLayout(new
BorderLayout(10, 10));

    Label title = new Label("STUDENT REGISTRATION FORM", Label.CENTER);
title.setFont(new Font("Arial", Font.BOLD, 20));    add(title,
BorderLayout.NORTH);

    Panel formPanel = new Panel();
formPanel.setLayout(new GridLayout(4, 2, 10, 10));

    idLabel = new Label("Student ID:");
idField = new TextField();

    nameLabel = new Label("Student Name:"); nameField
= new TextField();

```

```
genderLabel = new Label("Gender:");
```

```
genderGroup = new CheckboxGroup();    male = new  
Checkbox("Male", genderGroup, false);    female = new  
Checkbox("Female", genderGroup, false);
```

```
Panel genderPanel = new Panel(new FlowLayout(FlowLayout.LEFT));  
genderPanel.add(male);    genderPanel.add(female);
```

```
deptLabel = new Label("Department:");
```

```
department = new Choice();  
department.add("Select Department");  
department.add("Information Technology");  
department.add("Computer Science");  
department.add("Artificial Intelligence");  
department.add("Electronics and Communication");  
department.add("Mechanical Engineering");
```

```
formPanel.add(idLabel);  
formPanel.add(idField);
```

```
formPanel.add(nameLabel);  
formPanel.add(nameField);  
formPanel.add(genderLabel);
```

```
formPanel.add(genderPanel);
```

```
formPanel.add(deptLabel);
```

```
formPanel.add(department);
```

```
add(formPanel, BorderLayout.CENTER);
```

```
Panel bottomPanel = new Panel(new BorderLayout());
```

```
Panel buttonPanel = new Panel();
```

```
registerButton = new Button("Register");
```

```
clearButton = new Button("Clear");
```

```
registerButton.addActionListener(this);
```

```
clearButton.addActionListener(this);
```

```
buttonPanel.add(registerButton);
```

```
buttonPanel.add(clearButton);
```

```
bottomPanel.add(buttonPanel, BorderLayout.NORTH);
```

```
resultArea = new TextArea("", 8, 50);
```

```
resultArea.setEditable(false);
```

```
bottomPanel.add(resultArea, BorderLayout.CENTER);
```

```
add(bottomPanel, BorderLayout.SOUTH);
```

```
addWindowListener(new WindowAdapter() {  
    public void windowClosing(WindowEvent e) {  
        System.exit(0);  
    }  
});
```

```
setVisible(true);  
}
```

```
public void actionPerformed(ActionEvent e) {  
  
    if (e.getSource() == registerButton) {  
        registerStudent();  
    }  
  
    if (e.getSource() == clearButton) {  
        clearFields();  
    }  
}
```

```
private void registerStudent() {  
    String studentId = idField.getText().trim();  
    String studentName = nameField.getText().trim();  
  
    if (studentId.isEmpty() || studentName.isEmpty()) {  
        JOptionPane.showMessageDialog(this, "Please enter a valid ID and name.", "Error", JOptionPane.ERROR_MESSAGE);  
        return;  
    }  
  
    // Add student to the list  
    studentList.add(new Student(studentId, studentName));  
    updateTable();  
}
```

```
        if (studentId.isEmpty()) {  
resultArea.setText("Student ID is required");  
return;  
        }
```

```
        if (studentName.isEmpty()) {  
resultArea.setText("Student Name is required");  
return;  
        }
```

```
        Checkbox selectedGender = genderGroup.getSelectedCheckbox();
```

```
        if (selectedGender == null) {  
resultArea.setText("Gender is required");        return;  
        }
```

```
        if (department.getSelectedIndex() == 0) {  
resultArea.setText("Department is required");  
return;  
        }
```

```
        String gender = selectedGender.getLabel();  
        String dept = department.getSelectedItem();
```

```
        resultArea.setText(
```

```
        "Student Registered Successfully\n\n" +  
        "Student ID : " + studentId + "\n" +  
        "Name      : " + studentName + "\n" +  
        "Gender    : " + gender + "\n" +  
        "Department : " + dept  
    );  
}
```

```
private void clearFields() {
```

```
    idField.setText("");  
    nameField.setText("");  
    genderGroup.setSelectedCheckbox(null);  
    department.select(0);  
    resultArea.setText("");  
}
```

```
public static void main(String[] args) {  
    new StudentRegistration();  
}  
}
```

OUTPUT:

STUDENT REGISTRATION FORM

Student ID:

101

Student Name:

JESHWANT

Gender:

☒ Male ☐ Female

Department:

Information Technology

Register

Clear

Student Registered Successfully

Student ID : 101

Name : JESHWANT

Gender : Male

Department : Information Technology